Exercise 15.2

Solve by separation of variables, the following differential equations

1. $\frac{dy}{dx} = \frac{x}{y}$

Sol^n: $\frac{dy}{dx} = \frac{x}{y}$

$\Rightarrow \quad xdx - ydy = 0$

$\Rightarrow \quad \int x\,dx - \int y\,dy = 0$

$\Rightarrow \quad \frac{x^2}{2} - \frac{y^2}{2} = \frac{c}{2}$, where $\frac{c}{2}$ is arbitrary constant.

$\Rightarrow \quad x^2 - y^2 = c.$

2. $\mathbf{xdx + ydy = 0}$

Sol^n: $xdx + ydy = 0$

$\Rightarrow \quad xdx + ydy = 0$

$\Rightarrow \quad \int x\,dx + \int ydy = 0$

$\Rightarrow \quad \frac{x^2}{2} + \frac{y^2}{2} = \frac{c}{2}$, where $\frac{c}{2}$ is arbitrary constant.

$\Rightarrow \quad x^2 + y^2 = c.$

3. $\frac{dy}{dx} = \frac{x^2 + 1}{y^2 + 1}$

Sol^n: $\frac{dy}{dx} = \frac{x^2 + 1}{y^2 + 1}$

$\Rightarrow \quad (y^2 + 1)\,dy = (x^2 + 1)\,dx.$

$\Rightarrow \quad \int (y^2 + 1)\,dy = \int x^2 + 1\,dx$

$$\Rightarrow \quad \frac{y^3}{3} + y = \frac{x^3}{3} + x + c$$

$$\Rightarrow \quad \frac{x^3}{3} + x = \frac{y^3}{3} + y + c.$$

4. $\mathbf{x^2dy - y^2dx = 0}$

Sol^n: $x^2dy - y^2dx = 0$

Dividing by x^2y^2, we get

$$\frac{dy}{y^2} - \frac{dx}{x^2} = 0$$

Integrating, we get

$$\int \frac{dy}{y^2} - \int \frac{dx}{x^2} = 0 \qquad \Rightarrow -\frac{1}{y} + \frac{1}{x} = c \qquad \Rightarrow x - y = cxy.$$

5. $\mathbf{(1 + x^2)\frac{dy}{dx} = 1}$

Sol^n: $(1 + x^2)\frac{dy}{dx} = 1$

$$\Rightarrow \quad dy = \frac{1}{x^2 + 1}\, dx$$

Integrating, we get

$$\Rightarrow \quad \int dy = \int \frac{dx}{1+x^2} \qquad \Rightarrow y = \tan^{-1} x + c.$$

6. $\mathbf{\frac{dy}{dx} = \frac{e^x + 1}{y}}$

Sol^n: $\frac{dy}{dx} = \frac{e^x + 1}{y}$

$$\Rightarrow \quad ydy = (e^x + 1)\, dx$$

Integrating, we get

$$\int ydy = \int e^x dx$$

$$\Rightarrow \quad \frac{y^2}{2} = (e^x + x) + \frac{c}{2} \qquad \Rightarrow y^2 = 2e^x + 2x + c.$$

7. $\mathbf{\frac{dy}{dx} + 4x = 2.e^{2x}}$

Sol^n: $\frac{dy}{dx} + 4x = 2.e^{2x}$

$$\Rightarrow \quad dy + 4x.dx = 2.e^{2x}.dx$$

Integrating, we get

$$\int dy + 4\int x.dx = 2.\int e^{2x} dx$$

$$\Rightarrow \quad y + 4\frac{x^2}{2} = 2\,\frac{e^{2x}}{2} + c \qquad \Rightarrow y = e^{ex} - 2x^2 + c.$$

8. $\mathbf{\sqrt{1 - x^2}\, dy + \sqrt{1 - y^2}\, dx = 0}$

Sol^n: Dividing by $\sqrt{1 - x^2}\sqrt{1 - y^2}$, we get

$$\frac{dy}{\sqrt{1 - y^2}} + \frac{dx}{\sqrt{1 - x^2}} = 0$$

$$\Rightarrow \quad \int \frac{dy}{\sqrt{1 - y^2}} + \int \frac{dx}{\sqrt{1 - x^2}} = 0$$

Integrating, we get.

$$\sin^{-1}y + \sin^{-1}x = \sin^{-1} c$$

$$\Rightarrow \quad \sin^{-1}\{x\sqrt{1-y^2} + y\sqrt{1-x^2}\} = \sin^{-1}c$$

$$\Rightarrow \quad x\sqrt{1-y^2} + y\sqrt{1-x^2} = c.$$

9. $\mathbf{(1+x)\,ydx + (1+y)\,xdy = 0}$

Sol^n: Dividing both sides by xy, we get

$$\left(\frac{1+x}{x}\right)dx + \left(\frac{1+y}{y}\right)dy = 0$$

$$\Rightarrow \quad \frac{1}{x}dx + dx + \frac{1}{y}dy + dy = 0$$

Integrating, we get

$$\log x + x + \log y + y = c$$

$$\Rightarrow \quad (\log x + \log y) + (x + y) = c \quad \Rightarrow \log(xy) + x + y = c$$

10. $\mathbf{(xy^2 + x)\,dx + (yx^2 + y)\,dy = 0}$

Sol^n: $(xy^2 + x)\,dx + (yx^2 + y)\,dy = 0$

$$\Rightarrow \quad x(y^2+1)\,dx + y(x^2+1)\,dy = 0$$

Dividing by $(x^2+1)(y^2+1)$ we, get

$$\frac{xdx}{x^2+1} + \frac{ydy}{y^2+1} = 0$$

$$\Rightarrow \quad \frac{2xdx}{x^2+1} + \frac{2ydy}{y^2+1} = 0$$

Integrating, we get

$$\int\frac{2xdx}{x^2+1} + \int\frac{2ydy}{y^2+1} = 0$$

$$\Rightarrow \quad \log(x^2+1) + \log(y^2+1) = \log c$$

$$\Rightarrow \quad \log[(x^2+1).(y^2+1)] = \log c$$

$$\Rightarrow \quad (x^2+1)(y^2+1) = c.$$

11. $\mathbf{\frac{xdy}{dx} + y - 1 = 0}$

Sol^n: $\frac{xdy}{dx} + y - 1 = 0$

$$\Rightarrow \quad \frac{xdy}{dx} = -(y-1)$$

$$\Rightarrow \quad \frac{dy}{(y-1)} = \frac{-dx}{x}$$

Integrating, we get

$$\Rightarrow \quad \int\frac{dy}{(y-1)} = -\int\frac{dx}{x}$$

$$\Rightarrow \quad \log(y-1) = -\log(x) + \log(c)$$

$$\Rightarrow \quad \log(y-1) + \log(x) = \log(c)$$

$$\Rightarrow \quad \log\{(y-1)x\} = \log c \qquad \Rightarrow x(y-1) = c.$$

12. $\frac{dy}{dx} = e^x - y + x^3.e^{-y}$

Sol^n: $\frac{dy}{dx} = e^x.e^{-y} + x^3.\ e^{-y}$

$\Rightarrow \quad \frac{dy}{dx} = e^{-y}(e^x + x^3)$

$\Rightarrow \quad \frac{dy}{e^{-y}} = (e^x + x^3)\ dx$

Integrating, we get

$\int e^y dy = \int (e^x + x^3)\ dx$

$\Rightarrow \quad e^y = e^x + \frac{1}{4}x^4 + c.$

13. **tanx dy + tany dx = 0**

Sol^n: Dividing both sides by tan x.tan y, we get

$\frac{dy}{\tan x} + \frac{dy}{\tan y} = 0$

$\Rightarrow \quad \frac{\cos x}{\sin x}\ dx + \frac{\cos y}{\sin y} dy = 0$

Integrating, we get

log sin x + log siny = log c, where log c is arbitrary constant.

$\Rightarrow$ log (sinx. siny) = log c $\Rightarrow$ sinx siny = c.

14. $\frac{dy}{dx} = \frac{1 + \cos 2y}{1 - \cos 2x}$

Sol^n: $\frac{dy}{dx} = -\frac{1 + \cos 2y}{1 - \cos 2x}$

$\Rightarrow \quad \frac{dy}{1 + \cos 2y} + \frac{dx}{1 - \cos 2x} = 0$

$\Rightarrow \quad \frac{dy}{2\cos^2 y} + \frac{dx}{2\sin^2 x} = 0$

$\Rightarrow \quad \frac{dy}{\cos^2 y} + \frac{dx}{\sin^2 x} = 0$

$\Rightarrow \quad \sec^2 y\ dy + \text{cosec}^2\ x dx = 0$

Integrating, we get

tany – cotx = c, where c is arbitrary constant.

15. $x\frac{dy}{dx} + y = 2$

Sol^n: $\frac{dy}{dx} + y = 2$

$\Rightarrow \quad \frac{dy}{y - 2} = -\frac{dx}{x}$

Integrating, we get,

$\int \frac{dy}{y - 2} - \int \frac{dx}{x} = 0$

$\Rightarrow$ log (y – 2)= – log x + log c $\Rightarrow$ log(y – 2) + log x = log c

$\Rightarrow$ log {(y – 2)x} = logc $\Rightarrow$ x(y – 2) = c.

Hint and Solution of MCQ's

1. The linear differential equation is of the form $\frac{dy}{dx} + Py = Q$, where P and Q are function in x.

   $\therefore$ The equation $x\frac{dy}{dx} + 4y = x^2$

   $\Rightarrow \quad \frac{dy}{dx} + \frac{4}{x}y = x$ is linear.

2. The linear differential equation is first order differential equation of the form $\frac{dy}{dx} + Py = Q$ Where P and Q are function in x.

   $\therefore \quad y\frac{dy}{dx} + 5y = x \quad \Rightarrow \frac{dy}{dx} + 5 = \frac{x}{y}$ is not linear as $Q = \frac{x}{y}$ is not function in x

3. The highest order derivative in the equation is second order.

   $\therefore$ order of the equation is 2

4. Since, power of highest order is degree of the equation

   $\therefore$ degree of the equation is 2

5. Highest order derivative = 4, the order of equation
   power of highest order = 1, the degree of equation

6. $\left[1 + \left(\frac{dy}{dx}\right)^4\right]^{1/3} = \frac{d^2y}{dx^2}$

   $\Rightarrow \quad 1 + \left(\frac{dy}{dx}\right)^4 = \left(\frac{d^2y}{dx^2}\right)^3$

   $\therefore$ Highest order derivative = 2, the order of equation
   power of highest derivative = 1, the degree of equation

7. Highest order derivative = 3, the order of equation
   power of highest derivative = 1, the degree of equation
   and the differential equation is non-linear

8. $y = Ae^x$ ..... (i)

   $\frac{dy}{dx} = A.e^x$

   $\Rightarrow \quad A = e^{-x}.\frac{dy}{dx}$

   $\therefore$ from (i), $y = e^{-x}.\frac{dy}{dx}.e^x$

   $\Rightarrow \quad \frac{dy}{dx} = y \qquad \Rightarrow \frac{dy}{dx} - y = 0$

9. $y = 2\sin x + 3\cos x$ ..... (i)

   $\Rightarrow \quad \frac{dy}{dx} = 2\cos x - 3\sin x \qquad \Rightarrow \frac{d^2y}{dx^2} = -2\sin x - 3\cos x$

   $\Rightarrow \quad \frac{d^2y}{dx^2} = -(2\sin x + 3\cos x)$

   $\Rightarrow \quad \frac{d^2y}{dx^2} = -y \qquad \Rightarrow \frac{d^2y}{dx^2} + y = 0.$

10. The differential equation in first order and first degree may be in the form all of variable separable, homogeneous and linear.

11. The solution of the differential equation

$$\frac{dy}{dx} = \frac{f(x)}{g(y)}$$

$\Rightarrow$ $g(y)\,dy = f(x)\,dx$ $\Rightarrow \int g(y)\,dy = \int f(x)\,dx + c$

12. $\frac{dy}{dx} = 2x$ $\Rightarrow dy = 2xdx$

on integration,

$$\int dy = 2\int xdx$$

$\Rightarrow$ $y = x^2 + c$

we have $y(1) = 2$ $\Rightarrow y = 2$ when $x = 1$

then $2 = 1^2 + c$ $\Rightarrow c = 1$

$\therefore$ $y = x^2 + 1$ is the solution.

13. $\frac{dy}{dx} = -\frac{y}{x}$ $\Rightarrow \frac{dy}{y} = -\frac{dx}{x}$

$\Rightarrow$ $\int \frac{dy}{y} = -\int \frac{dx}{x}$ $\Rightarrow \log y = -\log x + \log c$

$\Rightarrow$ $\log y = \log \frac{c}{x}$ $\Rightarrow y = \frac{c}{x}$ $\Rightarrow xy = c$ .... (i)

We have $y(2) = 5$ i.e. $y = 5$ when $x = 2$

So, $xy = c$ $\Rightarrow 10 = c$

$\therefore$ from (i), the required solution is $xy = 10$

14. $\frac{dy}{dx} = e^{2x-3y}$ $\Rightarrow \frac{dy}{dx} = e^{2x}.\, e^{-3y}$

$\Rightarrow$ $\frac{dy}{e^{-3y}} = e^{2x}\,dx$ $\Rightarrow e^{3y}\,dy = e^{2x}\,dx$

On Integration

$$\frac{e^{3y}}{3} = \frac{e^{2x}}{2} + c$$

15. $\sqrt{1-x^2}\,dy - \sqrt{1-y^2}\,dx = 0$

$\Rightarrow$ $\mathrm{Sin}^{-1}y - \sin^{-1}x = C$

We have $y\left(\frac{1}{2}\right) = \frac{\sqrt{3}}{2}$ i.e. $y = \frac{\sqrt{3}}{2}$ when $x = \frac{1}{2}$

$$\sin^{-1}\left(\frac{\sqrt{3}}{2}\right) - \sin^{-1}\left(\frac{1}{2}\right) = C$$

$\Rightarrow$ $C = \frac{\pi}{3} - \frac{\pi}{6} = \frac{\pi}{6}$